

Viva Questions & Answers (Hinglish)

Topic: Kalman Filter, Trackers & Deep Learning Filters

Paper 1: Kalman Filter based Tracking for Fast■Moving Tiny Objects

Q: Is paper ka main goal kya hai?

A: Is paper ka goal yeh analyze karna hai ki existing Kalman Filter based trackers (jaise SORT, ByteTrack, StrongSORT) fast-moving tiny objects (racquetball) ko accurately track kyun nahi kar paate.

Q: Fast■moving tiny objects track karna difficult kyun hota hai?

A: Kyunki object chhota hota hai (kam pixels), speed zyada hoti hai, motion blur aata hai, aur direction suddenly change hoti hai — jo KF ke linear motion assumption ko tod deta hai.

Q: Is paper ka key conclusion kya hai?

A: Sabhi KF-based trackers mein large spatial error aata hai (3–11 cm), jo practical applications ke liye unacceptable hai. Matlab KF alone sufficient nahi hai.

Paper 2: Beyond Kalman Filters – Deep Learning Based Filters

Q: Is paper ne Kalman Filter ko kyun challenge kiya?

A: KF linear motion assume karta hai aur manual tuning maangta hai. Real-world motion non-linear hoti hai, jahan KF fail karta hai.

Q: Filter-level learning ka matlab kya hai?

A: Filter-level ka matlab hai sirf motion estimation aur noise correction part ko deep learning se replace karna — poora tracker nahi.

Q: End-to-end filter ka advantage kya hai?

A: Isme measurement noise, process noise manually define nahi karna padta. Model khud detector errors ko learn kar leta hai.

Q: Deep learning slow hone ke bawajood kyun use kiya?

A: Kyuki accuracy, robustness aur occlusion handling bahut improve hoti hai. Sports ya dance jaise complex motion mein speed se zyada accuracy important hoti hai.

Paper 3: Trackers using Kalman Filter (SORT family)

Q: Tracker-level aur filter-level mein difference kya hai?

A: Filter-level sirf state estimation karta hai (KF / DL filter). Tracker-level poora system hota hai — detection, association, track management.

Q: SORT ka basic working kya hai?

A: Har frame detection hoti hai, KF next position predict karta hai, IoU + Hungarian algorithm se matching hoti hai.

Q: High-IOU SORT / OC-SORT jaise trackers kya improve karte hain?

A: Ye occlusion handling, re-initialization aur better association logic add karte hain — but still KF use karte hain.

Q: Is paper se kya learning milti hai project ke liye?

A: Kalman Filter fast hai but complex motion mein limited. Agar high accuracy chahiye, to DL-based filters future direction hain.